

Designing Collaboration between Human Beings and Self-driving Heavy Vehicles with Emerging Interaction Technologies

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ABSTRACT

The present work-in-progress paper describes the development of novel user interface concepts that allow human operators to collaborate with self-driving heavy vehicles in a mining context. Concept development was performed within a user-centered design process containing three main steps. First, a study was performed to identify interaction points between heavy vehicle drivers and other human operators in mines. Second, potential interaction technologies were investigated. Finally, suggestions for interaction models were designed and implemented in 3D animated movies. The concepts were designed to support human operators performing loading tasks together with self-driving vehicles and utilize voice interaction and an augmented reality head-up-display to facilitate the interaction. In addition to the mining context, similar concepts were developed to support forklift drivers performing loading tasks in logistic centers. In the next step of this project, the suggested interaction models will be evaluated with mine workers and forklift drivers.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Interaction paradigms**; • **Collaborative interaction**;

KEYWORDS

Self-driving vehicles, Autonomous vehicles, Interaction design, Verbal interaction, Augmented reality, Artificial intelligence

ACM Reference Format:

Dr. Johan Fagerlön, Anna Sirkka, Lina Orrell, Dr. Yanqing Zhang, Stefan Larsson, Elin Tybring, and Hanna Rönntoft. 2021. Designing Collaboration between Human Beings and Self-driving Heavy Vehicles with Emerging Interaction Technologies. In *13th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI '21 Adjunct)*, September 09–14, 2021, Leeds, United Kingdom. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3473682.3480278>

1 INTRODUCTION

The implementation of self-driving vehicles (SDVs) suggests great potential to increase safety, efficiency, and to reduce environmental harm. The ongoing innovation project In the hub aims to explore and demonstrate new models of interaction between professional human operators and self-driving heavy vehicles (SAE automation level 4-5, [1]).

The present work focuses primarily on exploring new solutions for the mining industry. In this domain, highly automated heavy vehicles have already been implemented to some extent, and there is a need to understand how new technology may be implemented to allow appropriate human-vehicle interaction. An especially interesting situation is when operators must work in the close vicinity of vehicles. These situations arise when operators and automated vehicles share spaces and need to collaborate on common tasks. Here, new interaction models should be developed to ensure high levels of safety, efficiency, acceptance, and trust towards the automation.

Previous research has investigated how new interaction models can replace human interaction when humans share spaces with SDVs. For instance, studies have examined the usefulness of external interfaces, to facilitate safety and comfort [2–4]. However, there is a need to better understand how interaction models can be designed to handle the complex interaction needed when operators work together with the vehicles. The present paper describes the development of novel user interface concepts allowing operators to collaborate with self-driving heavy vehicles in a mining context.

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AutomotiveUI '21 Adjunct, September 09–14, 2021, Leeds, United Kingdom

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ACM ISBN 978-1-4503-8641-8/21/09.

<https://doi.org/10.1145/3473682.3480278>

2 CONCEPT DESIGN

The concept development was performed within a design process containing three main steps. First, a study was performed to better understand how collaboration between operators and heavy vehicle drivers is conducted, as well as identifying critical interaction points. Second, potential interaction technologies were investigated. Finally, suggestions for interaction models were designed.

2.1 Investigating interaction points

Initially, as part of the design process, user journey mapping was conducted. The mapping was performed with 13 mine workers (loader operators, heavy-vehicle drivers etc.) working in three different underground mines. The objective of the mapping was to better understand the mine worker’s behavior. Primarily, the study aimed to better understand how collaboration between operators and heavy vehicle drivers is conducted, as well as identifying critical interaction points.

A critical interaction point refers to an interaction between an operator and a driver where it is not obvious how the interaction will take place in a future where the drivers are no longer present, and the interaction will be between an operator and an SDV.

The mapping was conducted through semi-structured interviews. A typical working day was visualized by plotting activities against a timeline where the starting point was the beginning of the working day. Interaction points, with humans and systems, were highlighted.

The information gathered was then clustered and analyzed. The main purpose was to find frequently occurring interaction points and situations where the communication between an operator and a SDV is of high importance. Scenarios were then developed that would later set basis for the concept development.

Among the interaction points identified, the loading task was identified as a situation that often requires collaboration and several steps of communication between the truck driver and the loader operator. Furthermore, loading is strongly connected to production efficiency. Thus, it was decided to develop concepts on supporting interaction during loading (a heavy vehicle transporting rocks, SAE automation level 4-5 [1]). This involves preparation before the loading takes place (e.g., how the vehicle should position itself in relation to the loader), the actual loading (e.g., weight distribution), and load securing before transportation. The results of the study also showed that operators today use a variety of available methods to communicate before and during the loading, such as using the radio, the horn and the headlights. For instance, they may use verbal communication to determine how the truck should be positioned in the loading area, while e.g. the horn may be used to indicate that the vehicle should stop.

2.2 Interaction technologies

There are several potential technologies that could be suitable for interaction with SDVs including, for example, gesture recognition, tablets, and physical controllers. Research has suggested how verbal communication may facilitate interaction between self-driving cars and passengers [5, 6], as well as pedestrians near the vehicle [6]. In the present work, verbal interaction was identified as particularly interesting to explore for three main reasons: 1. The interviews with mine workers indicate that verbal interaction is a common

way to communicate (by using the radio); thus, the interaction modality may allow the users to employ common terminology to interact with the vehicle. 2. Verbal interaction allows the user to communicate while keeping the hands and eyes free. 3. Assigning human-like features to the autonomous vehicle and its interface, referred to as anthropomorphism, has been shown to increase trust and likeability. Waytz, Heafner and Epley [7] gave their car a voice, a name, and a gender in a simulator study. They found that overall trust and likeability was higher with these additions. Similarly, Ruijten, Terken and Chandramouli [8] showed that conversational interfaces result in higher trust compared to a traditional Graphical User Interface. Research has also indicated that a more social and conversational interface may contribute to higher levels of trust and pleasantness compared to using voice commands [4].

A challenge associated with verbal interaction is that users may need to learn a set of predefined commands. However, the ongoing development of AI-supported natural language understanding services (e.g., Microsoft LUIS and IBM Watson) indicates that future implementation may offer an experience very similar to natural conversations with humans.

Today, system information in loaders is typically presented on screens and tablets. However, looking at screens while operating a machine could become distracting and a safety issue. The use of Head-up Displays (HUD) and Augmented Reality (AR) have been investigated to support operators in various domains, including the mining industry [9]. Overall, the research indicates that the technology can improve operators’ productivity, confidence, and safety [10]. For instance, Wallmyr, Taufik and Sitompul [11] showed that placing information near the line of sight may reduce cognitive workload compared to using traditional “look-down” displays. AR also allows information to be placed virtually on objects in the environment, which may allow the operator to quickly access and interpret the information. Further, AR allows the operator to “see through” obstacles and potentially gain better situational awareness. Thus, a HUD solution with AR capabilities, in addition to verbal interaction, was identified as an especially interesting technology to explore.

An overall challenge with HUDs and AR is that the information may obscure the real world. Thus, the placement and design of visualizations must take this into account; for instance, the use of semi-transparent AR has been suggested [12]. Another challenge is that visual information on the AR HUD may be challenging to use in bright environments. However, this makes the damped light conditions of underground mines especially interesting for AR technology.

Combining verbal interaction and AR is not new. For instance, Microsoft HoloLens offers both interaction modalities and the combination has for instance been suggested for interaction with loader cranes [13]. However, there is little known how these upcoming technologies could be combined to support collaboration with SDVs.

Table 1: Distribution of communication and information in the NVI concept.

Verbal interaction	AR HUD
Commands. For example, how the vehicle should position itself in relation to the loader. Proactive questions. For example, the vehicle asks if it should lock itself to prevent other human operators from taking control of the vehicle during the loading. Social communication. For example, the vehicle greets the operator and provides encouraging feedback during the loading task. The vehicle also calls the human operator by name. Warnings. For example, that the load is inappropriately distributed in the vehicle.	Real time information about vehicle intentions. For example, showing how the vehicle intends to reverse and stop inside the loading area. Real time information about the load. For example, how the load is distributed in the vehicle. Information to support working memory. For example, the name of the vehicle and when it will arrive to the loading area. Warnings. For example, that the load is inappropriately distributed in the vehicle.

2.3 Interaction model design

The design of interaction models was conducted in an iterative process involving a 12-person team of designers, researchers, 3D-animators and people working in the mining industry. Each iteration consisted of an approx. one-week design phase, followed by a workshop where the team members reviewed the concepts and provided feedback and suggestions for improvements. In the final iterations, the concepts were implemented in 3D animated movies. The movies show a loading scenario from the first-person perspective in which the operator interacts with the vehicle to prepare, conduct, and finalize a loading task. The voice of the vehicle was created by using the Microsoft Azure text-to-speech service. The operator voice was made by an actor.

The process resulted in two concepts: Natural Voice Interaction (NVI) and Basic Voice Interaction (BVI). The main difference between the concepts lies in the way voice interaction was implemented. Table 1 shows how communication and information was distributed between the interaction modalities in the NVI concept. The concept was designed to mimic natural and social verbal interaction between two human operators performing the task together. The design of the BVI concept shares many similarities with the NVI concept. However, as opposed to the NVI concept, the use of verbal interaction was kept to a minimum. This means:

- No social communication between the human operator and the vehicle. The exchanged information is only related to the loading task. The vehicle does not call the operator by name.
- Messages from the vehicle are presented as short written notifications in the HUD or brief verbal acknowledgements and feedback (e.g., “OK”).
- The operator only uses short commands to control the vehicle (e.g., “Lock vehicle”).
- The vehicle does not ask the operator any questions.

Figure 1 shows an example of how visual information was presented in both concepts. In this example, the operator is about to load a bucket of rocks into the compartment of the truck. The HUD uses a combination of 2D and 3D (projected on objects) visualizations. The 2D visualization bar shows the load capacity and current load in the bucket, as well as information about the vehicle and the loading area. The 3D AR visualizations are used to provide information

about how the current load is distributed in the vehicle by allowing the user to “see through” the load compartment of the vehicle. The created 3D animated videos describing the full loading scenario for the concepts are available online, see [14] for the NVI concept and [15] for the BVI concept.

3 NEXT STEP

A study will be conducted during 2021 to evaluate the concepts with mine workers. One aim is to gather insights on how the interaction models can affect acceptance, efficiency, safety, and trust towards the automated vehicle. The study will focus particularly on investigating the potential and limitations of using natural voice interaction and social interaction. A second aim is to gather input on how the interaction models could be improved before being implemented as a prototype and evaluated in a real user context. The study will be conducted using interviews. The subjects will watch the movies presenting the concepts and options will mainly be assessed using rating scales and open questions.

Supporting operators in the mining industry is a limited area of application. Thus, corresponding NVI and BVI concepts have also been developed to support forklift drivers working in logistic centers. Facilitating forklift drivers using AR has shown promising results [12], but there is little known about how to support collaboration with SDVs in this context. Figure 2 shows an example of how information is presented to the forklift operator. In this example, the vehicle guides the forklift driver where to put the next pallet (using a virtual green pallet). These concepts will also be evaluated during 2021. Taken together, the studies will provide better insights into how the suggested interaction models could be applied to different types of environments, and how they need to be adapted to different types of operators.

4 CONCLUSION

The present work describes the development and design of novel concepts of how interaction and collaboration between loader operators and self-driving heavy vehicles could be facilitated using a combination of verbal interaction and an AR HUD. Natural language understanding services and full-view AR HUDs are two upcoming technologies that open up new possibilities for future interaction design. The concepts created suggest how the technologies may be



Figure 1: Information presented in the AR HUD during loading. The bar to the left in the 2D visualization shows current load weight in the truck (blue) and load weight in the bucket (white). The position of the loading task in the mine is L810 and the name of the vehicle is AV7. The lock icon indicates that the SDV can only be operated by the loader operator. Load distribution is presented as 3D visualizations in the load compartment.



Figure 2: Information presented in the AR HUD during loading in a logistic center.

used in future implementations. User evaluations conducted during 2021 with mine workers and forklift drivers will examine whether the interaction models are suitable in real user contexts.

ACKNOWLEDGMENTS

The work was funded by the Swedish partnership program Strategic Vehicle Research and Innovation (FFI) (Grant no: 2019-05898). It was

conducted in collaboration with Scania, Boliden Mineral, Icemakers and AFRY.

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